TraVIS: Combining Player Tracking and Video for Football Action Spotting

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Abstract

Action spotting aims to localize actions in long, untrimmed videos using a single keyframe per action, making it well suited to the fast-paced nature of sports. Focusing on football, we identify two remaining challenges. First, although existing methods account for different temporal context lengths and scales, their relative effectiveness has not been systematically compared. Second, most existing methods rely exclusively on video features, requiring models to infer football-specific spatial and relational structure implicitly from appearance and motion, which is particularly challenging when only a limited number of annotated matches are available. We address these limitations through two contributions. First, we conduct a comparative analysis of single- and multi-scale temporal architectures, showing that multi-scale models provide only modest gains over single-scale alternatives in video-only settings relative to their additional computational cost. Second, we introduce TraVIS, an action spotting framework that complements video features with explicit player and ball tracking information. TraVIS represents the game state as a graph whose nodes encode players and the ball and whose edges capture their spatial and relational structure, thereby introducing football-specific inductive biases and explicitly representing information that video-only approaches must otherwise learn implicitly. Fusing the graph and video representations achieves the strongest performance among the evaluated configurations on a proprietary football dataset, improving over the video-only model by +6.42 mAP@1s. These results show that structured tracking information provides a valuable complement to video features for football action spotting.

CCS Concepts

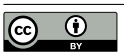
• **Computing methodologies** → **Activity recognition and understanding**; *Video segmentation*; Neural networks.

Keywords

Action Spotting, Player Tracking, Multimodal Fusion, Graph Neural Networks, Football Video Analysis

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1 Introduction

Action spotting aims to localize the frame at which actions occur in long, untrimmed videos. Unlike temporal action localization, which predicts start–end intervals, action spotting assigns each action to a single instant. This formulation is well suited to fast-paced sports, where action boundaries are often ambiguous, but can be more naturally represented by a characteristic keyframe [8]. The task supports practical downstream applications such as automatic match indexing, tactical analysis, and performance analytics.

In football, most prior work on action spotting has focused on broadcast footage under the SoccerNet benchmark [7, 8], where partial field coverage, camera cuts, and replays limit observability and disrupt temporal continuity. In this setting, E2E-Spot [12] demonstrated that a simple end-to-end architecture provides a strong baseline, with subsequent work exploring more sophisticated temporal modeling [18, 24] and multi-resolution approaches [25]. Despite these advances, two main axes remain understudied. First, although several temporal architectures have been proposed, the influence of temporal context length and scale has received limited systematic analysis. Different actions may require different amounts of temporal context to be correctly identified, while the spotting task still requires accurate localization of a single timestamp regardless of how much context was needed to establish it. Second, most existing methods rely exclusively on video input (Fig. 1a) and must therefore infer football-specific structure, such as player positions, team organization, and ball dynamics, implicitly from appearance and motion. Player and ball tracking data (Fig. 1b) provides this structured information directly, introducing spatial and relational inductive biases that can complement the visual stream.

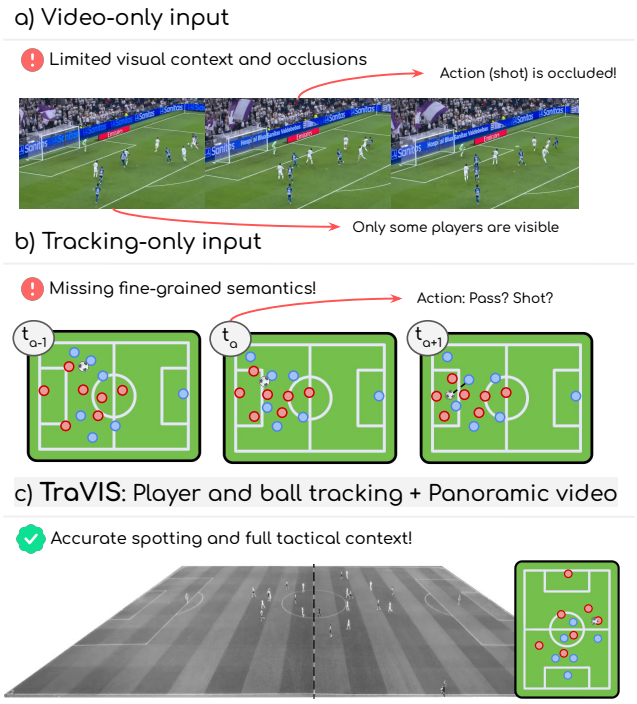


Figure 1: Illustration of standard football action spotting approaches: (a) video-only inputs suffer from limited visual context and severe occlusions, often missing players outside the frame; whereas (b) tracking-only inputs capture the global tactical layout but lack the fine-grained visual semantics crucial for distinguishing specific actions (e.g., a pass versus a shot). In contrast, (c) TraVIS fuses panoramic video—shown in grayscale for anonymization purposes only— with 2D player and ball tracking, effectively combining precise visual cues with full tactical context for precise action spotting.

These two gaps motivate our study on *PanoTrack*, a proprietary dataset of full professional football matches recorded using fixed wide-angle panoramic cameras, providing continuous full-pitch coverage. The dataset also includes synchronized player and ball tracking data and annotations for ten ball-related action classes. Using this setting, we address the two challenges by systematically comparing single- and multi-scale temporal modeling approaches and by introducing TraVIS, an action spotting framework that integrates structured tracking information with visual features (Fig. 1c). Our contributions can be summarized as follows:

(1) We conduct a systematic comparison of temporal modeling strategies for football action spotting, covering single- and multi-scale approaches based on GRUs, TCNs, and Transformer encoders. Our analysis shows that a simple single-layer bidirectional GRU provides a strong baseline in this fast-action setting, while multi-scale modeling offers only modest gains relative to its additional complexity.
(2) We introduce TraVIS, a multimodal action spotting framework that complements visual features with graph-based representations of player and ball tracking data. By explicitly encoding the game state, the proposed approach introduces football-specific spatial and relational inductive biases that would otherwise need to be inferred from visual data alone.
(3) Through extensive experiments and ablations, we show that visual and tracking information provide complementary cues, with TraVIS improving over the vision-only baseline by +6.42 mAP@1s points. We further analyze design choices required for effective multimodal integration, including auxiliary supervision, temporal modeling within the tracking stream, fusion strategies, and graph construction.

2 Related Work

Action Spotting (AS) aims to locate actions in untrimmed videos by assigning each occurrence a single timestamp, rather than estimating its start and end times as in Temporal Action Localization (TAL) [3, 15]. This formulation is particularly well suited to sports, where action boundaries may be ambiguous, while the event itself can often be represented more naturally by a characteristic instant grounded in the rules of the game [8, 10]. In football, research has largely developed around the SoccerNet benchmark [7, 8], with subsequent advances summarized in [9]. SoccerNet is based on broadcast footage and has been studied under two main annotation settings: a sparse setting covering high-level match events and a denser setting focused on ball-related actions. In contrast, we study *PanoTrack*, a proprietary dataset comprising two synchronized wide-angle panoramic views that provide continuous full-pitch coverage, with dense annotations of ball-related actions.

Temporal modeling for AS. Temporal approaches for AS can be characterized along two largely orthogonal dimensions: how the model is trained and how temporal context is represented.

The first dimension concerns the training regime. Early pipelines performed temporal reasoning over pre-extracted features while keeping the visual backbone frozen [5, 8, 11, 20, 23]. In contrast, E2E-Spot [12] demonstrated that a simple end-to-end architecture, jointly optimizing visual feature extraction and temporal modeling for the spotting objective, could achieve strong performance and paved the way for subsequent Action Spotting methods [18, 24, 25].

The second dimension concerns the temporal resolution at which context is modeled. *(a) Single-scale architectures* preserve a fixed temporal resolution and apply a temporal processor directly to the frame-level sequence, using models such as BiGRUs [4], which capture sequential dependencies using both past and future context; TCNs [1], which aggregate local evidence through convolutions with a controlled receptive field; and Transformer encoders [22], which model content-dependent interactions through bidirectional self-attention. *(b) Multi-scale architectures* instead operate at multiple temporal resolutions to combine fine-grained evidence with broader contextual information. In TAL, feature-pyramid methods such as TriDet [19] and ActionFormer [27] predict actions at different scales to account for variations in their temporal extent. This strategy is less directly applicable to precise AS, as predictions at coarse resolutions may compromise temporal localization. T-DEED [24] addresses this issue with an encoder-decoder architecture that progressively compresses the temporal dimension before restoring the original resolution through skip connections.

Following recent trends, we adopt an end-to-end framework and systematically compare single- and multi-scale architectures.

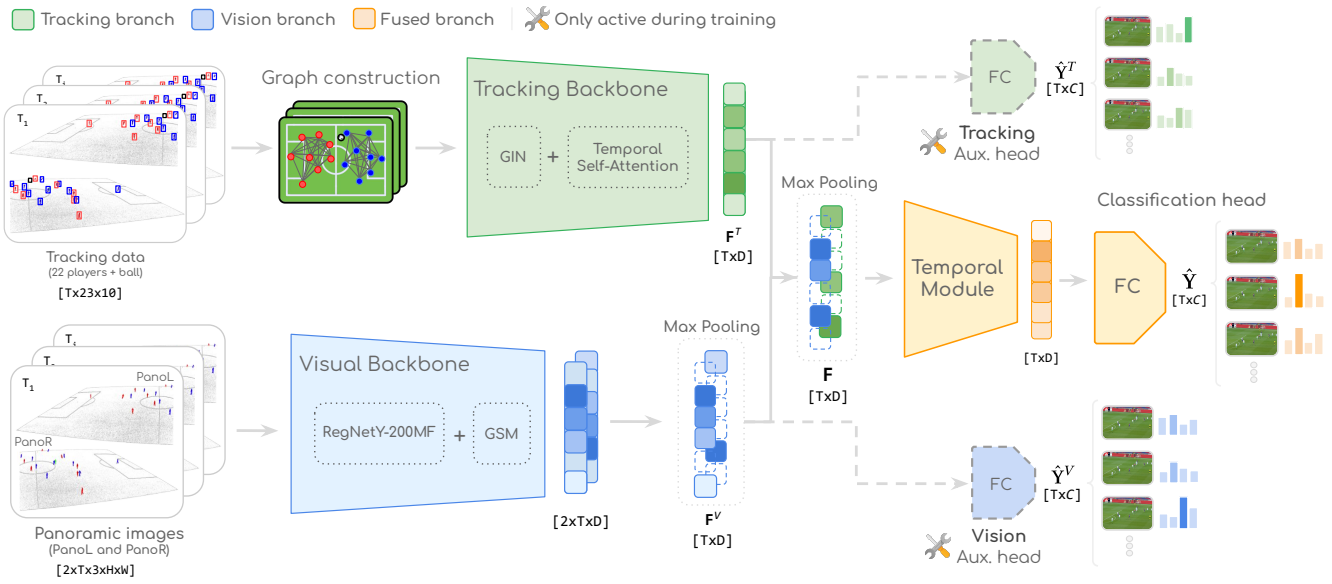


Figure 2: Overview of the proposed TRAVIS architecture. The vision branch (RegNetY-200MF + GSM, with view-level max-pooling) extracts visual features F^V, while the tracking branch (GIN-encoded per-frame graphs followed by temporal self-attention) extracts tracking features F^T. Both modalities are fused by element-wise maximum to form a joint multimodal representation F. This is processed by a Temporal Module to produce a context-enriched sequence, which a fully connected (FC) classification head maps to per-frame class probabilities. During training, two auxiliary FC heads compute modality-specific predictions ($\hat{Y}^T$ and $\hat{Y}^V$) to enforce that each branch preserves an independently discriminative representation. These auxiliary branches are discarded at inference.

Tracking data in sports analytics. Most AS methods seek to improve performance through increasingly sophisticated temporal modeling of visual features. However, video-only approaches must learn football-specific structure –such as player organization, spatial interactions, and ball dynamics– implicitly from appearance and motion. Tracking data provides this information explicitly through pitch-referenced player and ball trajectories, introducing domain-specific spatial and relational inductive biases that can complement the visual stream. Its potential has recently been demonstrated in football Group Activity Recognition, where Karki et al. [13] showed that a tracking-based model could outperform a strong video-only baseline while using substantially fewer parameters and less computation. Because tracking data naturally represents players and the ball as interaction entities, Graph Neural Networks (GNNs) [6] provide a suitable framework for modeling their relationships.

Building on these findings, we incorporate tracking data into our framework and model interactions among players and the ball using a Graph Isomorphism Network (GIN) [26].

3 Methodology

In this section, we first formulate the action spotting task and provide an overview of the proposed TRAVIS framework. We then describe its vision branch (Sec. 3.1), tracking branch (Sec. 3.2), multimodal fusion mechanism (Sec. 3.3), temporal module (Sec. 3.4), and prediction head (Sec. 3.5), followed by the training (Sec. 3.6) and inference (Sec. 3.7) details.

Problem formulation. Given an untrimmed video $\mathcal{V}$, Action Spotting aims to identify and temporally localize every action occurring in the video. The set of ground-truth action instances is denoted by $A = \{a_i\}_{i=1}^{N_a}$, where N_a is the number of annotated actions in $\mathcal{V}$ and may vary from one video to another. Each action instance $a_i = (t_i^*, c_i^*)$ is represented by a class label $c_i^* \in \{1, \dots, C\}$ and a timestamp t_i^* indicating the frame that best represents the occurrence of the action.

Method overview. Our proposed method TRAVIS, illustrated in Fig. 2, is designed for Action Spotting by complementing appearance and motion cues from visual features with spatial and relational inductive biases derived from player and ball tracking data. Videos are divided into fixed-length clips of T densely sampled frames for processing. First, a *vision branch* processes the frames from both panoramic views and fuses them at the feature level to produce per-frame visual features $\mathbf{F}^V$ (Sec. 3.1). In parallel, a *tracking branch* represents the corresponding tracking data as player-ball graphs and processes them with a GNN-based module to produce tracking features $\mathbf{F}^T$, matched to $\mathbf{F}^V$ in temporal length and dimensionality (Sec. 3.2). The two branches are then combined through element-wise maximum in a *multimodal fusion* stage (Sec. 3.3), after which a *temporal module* further enriches the fused features with temporal context (Sec. 3.4). Finally, a *classification head* maps each per-frame representation to a vector of class probabilities. During training, auxiliary prediction heads are attached to the visual and tracking branches immediately before fusion, encouraging each modality to learn independently discriminative representations and mitigating modality collapse during joint optimization.

3.1 Vision branch

The vision branch processes synchronized temporal sequences from the left and right panoramic views, denoted by $X_L, X_R \in \mathbb{R}^{T\times W\times H\times 3}$.

Each raw frame contains a central pitch region bordered by upper and lower bands depicting the stands and advertising boards, which provide limited information for action spotting. We therefore apply a fixed vertical crop before network processing, yielding $X_L^{\text{crop}}, X_R^{\text{crop}} \in \mathbb{R}^{T\times W\times H'\times 3}$, where $H' < H$. This operation preserves the pitch-visible region while reducing computational cost. The cropped sequences are then processed in parallel by a shared lightweight 2D convolutional backbone, RegNetY-200MF [17], producing per-view visual features $\mathbf{F}^L, \mathbf{F}^R \in \mathbb{R}^{T\times d}$. The backbone is initialized with ImageNet-1K pretrained weights and fine-tuned end-to-end for the action spotting task. Following E2E-Spot [12], we augment the backbone with Gated Shift Modules (GSM) [21] to capture short-range temporal information within the vision branch.

Multi-view fusion. The per-view representations are fused channel-wise by element-wise maximum, $\mathbf{F}^V = \max\left(\mathbf{F}^L, \mathbf{F}^R\right)$, $\mathbf{F}^V \in \mathbb{R}^{T\times d}$. Using a shared backbone encourages a common feature space across both views and reduces the number of trainable parameters relative to independent encoders, while max fusion retains the strongest activation from either view for each frame and feature channel.

3.2 Tracking branch

The tracking branch comprises three stages: graph construction, spatial-relational encoding, and temporal feature contextualization.

Player-ball graph construction. Player and ball tracking data naturally lends itself to a graph-based representation. At each frame t, we construct an attributed graph $\mathcal{G}_t = (\mathcal{V}, \mathcal{E}_t)$ with $|\mathcal{V}| = 23$ nodes, corresponding to the 22 players and the ball. Each node is described by a 10-dimensional feature vector comprising its normalized pitch coordinates (x, y), frame-to-frame displacement $(\Delta x, \Delta y)$, speed, acceleration, home- and away-team indicators, a tactical role index distinguishing goalkeepers, defenders, midfielders, and attackers, and a binary ball indicator. Rather than using a fully connected graph, which introduces many potentially uninformative relationships, we construct a sparse edge set using football-specific criteria. Specifically, $\mathcal{E}_t$ is the union of three complementary edge types (see Fig. 3). First, *ball-interaction edges* connect the ball node to every player, allowing information about its position and dynamics to propagate throughout the graph. Second, *tactical edges* connect players from the same team whose roles are identical or adjacent in the tactical hierarchy, such as goalkeeper–defender, defender–midfielder, and midfielder–attacker pairs. Finally, *proximity edges* connect any pair of nodes whose pitch distance at frame t is below a fixed threshold, capturing local interactions between nearby agents regardless of their team or tactical role. Together, these connections encode complementary ball-related, organizational, and spatial relationships while avoiding the computational and representational redundancy of full connectivity.

Spatial-relational graph encoding. Each frame-level graph $\mathcal{G}_t$ is processed by a Graph Isomorphism Network (GIN) [26]. GIN is well suited to this setting because it learns permutation-invariant representations while providing strong discriminative power over local graph structures. Through message passing, each node progressively aggregates information from players and the ball connected through tactical, proximity, and ball-interaction edges. At layer ℓ,

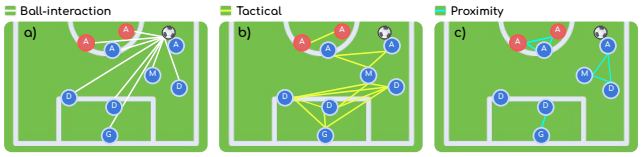


Figure 3: Illustration of the three graph-edge types used in our framework: a) *ball-interaction* edges, b) *tactical* edges, and c) *proximity* edges. For visual clarity, these are depicted in separate subgraphs, although they all coexist within a single unified graph. Only a subset of players is shown to simplify the visualization, with node letters indicating their tactical roles (G: Goalkeeper, D: Defender, M: Midfielder, A: Attacker).

the representation of node v is updated as

$$\mathbf{h}_v^{(\ell)} = \text{MLP}^{(\ell)}\left((1+\varepsilon^{(\ell)})\,\mathbf{h}_v^{(\ell-1)} + \sum_{u\in\mathcal{N}(v)} \mathbf{h}_u^{(\ell-1)}\right), \tag{1}$$

where $\mathcal{N}(v)$ denotes the neighborhood of node v, $\varepsilon^{(\ell)}$ is a learnable scalar, and $\text{MLP}^{(\ell)}$ is a two-layer perceptron with LayerNorm and ReLU activations. Each GIN layer is embedded in a residual `DeepGCNLayer` [14] to facilitate stable optimization. The resulting node embeddings have dimensionality d, matching that of the visual features and enabling element-wise multimodal fusion (Sec. 3.3). Finally, global mean pooling aggregates the node embeddings into a frame-level tracking representation $\mathbf{g}_t \in \mathbb{R}^d$. Frames without valid tracking information are represented by a zero vector.

Temporal feature contextualization. Because the GIN processes each frame independently, the resulting sequence $\mathbf{G} = \{\mathbf{g}_t\}_{t=1}^T$ captures spatial and relational structure but lacks temporal context. To align the tracking representation with the temporally enriched visual features, we process $\mathbf{G}$ using a lightweight temporal module comprising a single Transformer encoder layer [22] with learnable positional embeddings. This yields the final tracking features $\mathbf{F}^T \in \mathbb{R}^{T\times d}$, which encode both frame-level interactions and temporal dependencies and can be directly fused with the visual stream.

3.3 Multimodal fusion

We fuse the temporally aligned visual and tracking features, $\mathbf{F}^V$ and $\mathbf{F}^T$, by element-wise maximum, yielding $\mathbf{F} = \max(\mathbf{F}^V, \mathbf{F}^T) \in \mathbb{R}^{T\times d}$. This parameter-free operation combines appearance and motion cues from the vision branch with spatial and relational information from player and ball tracking, retaining the strongest response from either modality at each time step and channel. Max fusion is selected as it outperforms alternative strategies evaluated in Sec. 4.4.

3.4 Temporal module

In our comparative analysis, we consider two types of temporal modules for enriching the fused features $\mathbf{F}$ and producing the output representation $\mathbf{H}$: single-scale and multi-scale architectures.

Single-scale architectures. Single-scale architectures process the fused sequence $\mathbf{F}$ at its original temporal resolution, without temporal downsampling or upsampling. Under this setting, we evaluate stacked GRU [4], TCN [1], and Transformer encoder [22] layers, with the latter considering both local and global self-attention.

Multi-scale architectures. Multi-scale architectures process the fused sequence $\mathbf{F}$ at multiple temporal resolutions to capture dependencies over different ranges. Following [24], we use an encoder-decoder with S scales, progressive temporal downsampling by a factor κ, and skip connections to preserve fine-grained information. At each encoder scale, features are processed by N_l sequential layers before downsampling, followed by an additional sequential layer at the bottleneck. The decoder mirrors this process: at each scale, the features are upsampled by a factor κ using linear interpolation, combined by addition with the corresponding encoder features through a skip connection, and refined by N_l sequential layers. This process is repeated until the original temporal resolution is recovered. Based on the single-scale results reported in Sec. 4.4, we consider GRU [4] and TCN [1] layers as the sequential modules within this architecture.

Based on the comparative results in Sec. 4.4, we retain two temporal configurations for TraVIS: a single-scale bidirectional GRU and a multi-scale TCN-based encoder-decoder.

3.5 Prediction head

The prediction head outputs frame-level scores over the C action classes and an additional background class. A linear layer maps $\mathbf{H}$ to logits, which are converted by a softmax function into class probabilities $\hat{\mathbf{Y}} \in [0,1]^{T\times(C+1)}$.

3.6 Training details

Following Hong et al. [12], we formulate Action Spotting as a frame-wise classification problem. At each frame t, the model predicts a probability distribution $\hat{\mathbf{Y}}_t$ over the action and background classes. These predictions are supervised using the one-hot ground-truth labels $\mathbf{Y}$ and a weighted cross-entropy loss, $\mathcal{L}^w_{\text{CE}}(\mathbf{Y}, \hat{\mathbf{Y}})$, where w controls the relative contribution of foreground and background frames.

Auxiliary supervision. To mitigate modality collapse, we equip each branch with an independent auxiliary prediction pathway. Specifically, $\mathbf{F}^V$ and $\mathbf{F}^T$ are separately processed by a temporal module and a linear classification head that mirror the architecture of the main prediction branch, producing auxiliary predictions $\hat{\mathbf{Y}}^V$ and $\hat{\mathbf{Y}}^T$. These pathways are architecturally identical to, but do not share parameters with, the main branch. The overall objective is the average of the fused and modality-specific losses:

$$\mathcal{L} = \frac{1}{3}\left[\mathcal{L}^w_{\text{CE}}(\mathbf{Y}, \hat{\mathbf{Y}}) + \mathcal{L}^w_{\text{CE}}(\mathbf{Y}, \hat{\mathbf{Y}}^V) + \mathcal{L}^w_{\text{CE}}(\mathbf{Y}, \hat{\mathbf{Y}}^T)\right] \tag{2}$$

This auxiliary supervision encourages each modality to remain independently predictive, preventing the fused model from relying exclusively on a single branch and promoting the integration of complementary visual and tracking cues.

3.7 Inference

At inference time, the auxiliary prediction heads are removed and data augmentation is disabled. Each match half is processed using clips with 50% temporal overlap, and predictions for frames appearing in multiple clips are averaged. We then apply Soft Non-Maximum Suppression [2] to obtain discrete action spots.

4 Results

In this section, we describe the dataset (Sec. 4.1), evaluation protocol (Sec. 4.2), and implementation details (Sec. 4.3). We then present the ablation studies (Sec. 4.4) and evaluate the final performance of TraVIS, including a class-wise analysis of the results (Sec. 4.5).

4.1 Dataset

We conduct our experiments on *PanoTrack*, a proprietary multimodal dataset comprising 15 full football matches and approximately 22.5 hours of footage. All matches belong to the same professional league and correspond to home fixtures of a single team. Consequently, they were recorded at the same stadium using an identical camera setup. For each match, the dataset provides three temporally synchronized modalities: panoramic video, positional tracking data for all players and the ball, and keyframe-level action annotations.

Panoramic video recordings. Each match is captured by two fixed panoramic cameras on the same side of the pitch. As illustrated in Fig. 2, one camera covers the left half of the field (*PanoL*), while the other covers the right half (*PanoR*), jointly providing full-pitch coverage. Both cameras record at 25 fps in 4K UHD resolution. Unlike conventional broadcast footage, the recordings are continuous and do not contain camera cuts, replays, or viewpoint changes. This setup maintains full-pitch visibility throughout the match and avoids missing actions as a result of broadcast editing.

Positional tracking data. For each match, the dataset provides positional tracking for all players and the ball at 25 Hz, matching the video frame rate. Player and ball locations are represented by their x- and y-coordinates in meters within a pitch-centered coordinate system whose origin is located at the center of the field. Each entity is assigned a persistent identifier, enabling its trajectory to be followed throughout the match. Player metadata further includes team affiliation and role, while velocity and acceleration information is provided for all tracked entities.

Action annotations. Each match is annotated with the temporal locations and class labels of the target football actions. The dataset considers $C = 10$ action classes: *pass*, *reception*, *cross*, *corner*, *free-kick*, *goal-kick*, *throw-in*, *kick-off*, *tackle*, and *shot at goal*. Each match contains, on average, approximately 2,350 annotated actions. Tab. 5 reports the number of instances per class. The resulting class distribution is highly imbalanced: frequent actions such as *pass* and *reception* account for a large proportion of the annotations, whereas *kick-off*, *corner*, and *free-kick* occur considerably less often.

Therefore, the synchronized availability of the three data modalities enables the training and evaluation of the proposed TraVIS framework. The 15 matches are partitioned at the match level into 9 training, 3 validation, and 3 test matches, ensuring that no data from the same match appears in multiple splits. The same partition is used across all input modalities and experimental configurations.

4.2 Evaluation

Following standard practice, models are trained on the training split, with the validation split used for early stopping, and evaluated on the test split. Performance is measured using mean Average Precision at temporal tolerance δ (mAP@δs) [7], where a prediction

is considered correct if it falls within δ seconds of the corresponding ground-truth timestamp. We report results for $\delta \in \{1, 2\}$ seconds, denoted as mAP@1*s* and mAP@2*s*.

4.3 Implementation details

Unless otherwise stated, we train TraVIS on clips of $T = 50$ frames with a temporal stride of 4, corresponding to 8 seconds at 25 fps, using a batch size of 4 clips. Video frames are resized to $W \times H = 796 \times 448$ pixels and vertically cropped to $W \times H' = 796 \times 251$ pixels. Each epoch consists of 5,000 clips randomly sampled from the training split. Models are trained for 50 epochs using the AdamW optimizer [16] with an initial learning rate of 8e−4. The learning rate follows a linear warm-up during the first 10 epochs and subsequently follows a cosine decay schedule. Non-background classes are assigned a weight of $w = 5$ in the cross-entropy loss. All models use a hidden dimension of $d = 368$. The tracking self-attention block uses 4 attention heads, and for all encoder-decoder temporal modules, we use a temporal downsampling factor of $\kappa = 2$. During training, we apply standard data augmentations, including color jitter, Gaussian blur, horizontal flipping, and affine transformations to the input frames.

4.4 Ablation studies

We conduct the ablation studies sequentially, starting from a simplified, video-only version of TraVIS. The initial baseline uses a bidirectional GRU as the temporal module. We first evaluate the input configuration (i.e., vertical cropping, multi-view fusion, temporal stride, and clip length) in order to establish the default settings for subsequent experiments. We then compare a range of single- and multi-scale temporal modeling architectures. Finally, we incorporate the tracking stream and analyze the design choices involved in integrating and fusing the video and tracking representations.

4.4.1 Input configuration. Tab. 1 reports the input-configuration ablations. **(a) Vertical cropping.** The first experiment evaluates the vertical cropping strategy introduced in Sec. 3. Removing image regions outside the pitch, such as the stands and advertising boards, yields performance comparable to that obtained with full frames, indicating that these areas contribute little relevant information. At the same time, cropping reduces the frame height by approximately 44%, lowering computational and memory requirements without sacrificing accuracy and thereby supporting its use in subsequent experiments. **(b) Multi-view fusion strategy.** We compare the proposed parallel processing of the left and right views with horizontal concatenation, in which both views are combined into a double-width frame and processed jointly by the backbone. Horizontal concatenation reduces performance by 2.89 mAP@1s and 4.54 mAP@2s. Since football actions are often visible in only one view, element-wise max-pooling can more easily suppress an uninformative view while preserving the strongest responses from the informative one. **(c) Temporal stride.** Increasing the temporal stride extends clip coverage but reduces the sampling density. Among the evaluated settings, a stride of 4 achieves the best performance, sampling one every four frames and covering 8 seconds per clip. This configuration provides the best trade-off between capturing sufficient temporal context and preserving the temporal resolution required for accurate action spotting. **(d) Clip length.**

Table 1: Input-configuration ablations covering vertical cropping, multi-view fusion, temporal stride, and clip length. Temporal coverage is reported only for settings that alter the duration spanned by each clip. Bold configuration labels indicate the settings selected for subsequent experiments, while bold metric values denote the best result within each ablation.

Experiments	Coverage (s)	mAP@1s	mAP@2s
(a) Vertical cropping			
No	–	**65.84**	69.95
Yes	–	65.74	**70.27**
(b) Multi-view fusion strategy			
Parallel processing	–	**65.74**	**70.27**
Horizontal concatenation	–	62.85	65.73
(c) Temporal stride			
2	4	65.74	70.27
4	8	**68.30**	**71.57**
6	12	65.31	70.02
8	16	63.53	68.86
(d) Clip length			
T = 50	8	**68.30**	71.57
$T = 100$	16	66.90	**71.60**

Finally, we investigate whether longer clips provide useful additional temporal context. Increasing the clip length from $T = 50$ to $T = 100$ frames, thereby extending temporal coverage from 8 to 16 seconds, slightly decreases mAP@1s while producing comparable results on mAP@2s. These findings indicate that, under our architecture, the additional temporal context contributes little to action spotting. This may reflect the fast-paced nature of the actions, for which relatively local temporal information appears sufficient for reliable detection. We therefore retain $T = 50$, which also avoids the additional computational cost associated with processing longer clips.

4.4.2 Comparative analysis of temporal models. Tab. 2 compares alternative designs for the temporal modeling module. We evaluate both single- and multi-scale approaches using three representative sequential architectures: GRUs, TCNs, and Transformer encoders. We report the best-performing configuration for each. For GRUs, we vary the number of layers; for TCNs, we vary both the number of layers and the kernel sizes across layers; and for Transformer encoders, we vary the number of layers and the attention context. In the multi-scale setting, we retain the best-performing single-scale architectures and examine key design choices, including the number of temporal scales and the depth at each scale.

Single-scale analysis: **(a) Sequential architecture.** Under the single-scale setting, a single bidirectional GRU layer achieves the best performance, providing a strong yet simple choice for action spotting. Replacing the GRU with a TCN –whose best configuration uses four layers, a kernel size of 5, and exponentially increasing dilation– reduces mAP@1s by approximately 2.5 points. This suggests that the GRU captures both short- and long-range temporal dependencies more effectively in this setting. Transformer encoders perform substantially worse with both global and local attention. We observe unstable training behavior and high sensitivity to the learning rate, which we attribute in part to the limited size of the

Table 2: Comparison of single- and multi-scale temporal modeling configurations. We evaluate the sequential architecture, number of temporal scales, and depth at each scale. Best results within each experimental group are shown in bold.

Experiments		mAP@1s	mAP@2s
Single-scale	*(a) Sequential architecture*		
	GRU	**68.30**	**71.57**
	TCN	65.78	69.22
	TE (global attention)	59.84	66.84
	TE (local attention)	53.95	62.50
Multi-scale	*(a) Sequential architecture*		
	GRU	64.48	69.95
	TCN	**68.85**	**71.74**
	(b) Number of scales		
	S=2	**68.85**	**71.74**
	S=3	65.79	69.79
	S=4	63.71	69.45
	S=5	62.68	67.71
	(c) Scale depth		
	$N_l = 1$	**68.85**	**71.74**
	$N_l = 2$	63.31	68.55

training set. This is consistent with the general observation that attention-based sequence models often require more data, as they lack the strong locality priors inherent to recurrent and convolutional architectures. Given their poor performance, we exclude Transformer encoders from the multi-scale analysis.

Multi-scale analysis: **(a) Sequential architecture.** Unlike in the single-scale setting, TCNs outperform GRUs under multi-scale modeling, suggesting that multi-scale modeling is particularly well suited to TCNs, as it relaxes the constraints imposed by fixed kernel sizes and dilation schedules while enabling more effective modeling of temporal dependencies across different ranges. In contrast, GRUs already capture a broad temporal context and therefore benefit less from the multi-scale formulation. **(b) Number of scales.** Two temporal scales yield the best performance, while adding further scales provides no additional benefit. Coarser scales rely on increasingly downsampled temporal representations, which may sacrifice the temporal precision required for action spotting. The model appears unable to fully recover this lost information through the skip connections. **(c) Scale depth.** Increasing the number of layers per scale also brings no improvement, indicating that additional within-scale processing is unnecessary in this setting.

Overall, a single bidirectional GRU layer provides a remarkably strong baseline in our setting and represents the best-performing single-scale configuration. The multi-scale formulation, particularly when combined with TCN layers, yields only modest improvements of +0.55 mAP@1s and +0.17 mAP@2s over the single-scale GRU. We hypothesize that these limited gains arise from the relatively short temporal context required to recognize the actions considered in our setting. Because football actions are typically brief, local temporal information may already be sufficient to capture the relevant dynamics, thereby limiting the advantages of architectures designed to model longer-range dependencies. Multi-scale modeling may prove more beneficial in scenarios involving longer or more temporally complex actions. Moreover, these modest gains come at an additional computational cost, as the multi-scale model contains approximately 1.8× as many parameters as the single-scale GRU. Given this trade-off, we retain both configurations for the subsequent experiments on tracking integration.

Table 3: Comparison of TraVIS across temporal models. Best results are shown in bold.

Experiments	mAP@1s	mAP@2s
TraVIS (single-scale GRU)	**75.27**	**77.35**
TraVIS (multi-scale TCN)	70.99	72.68

Table 4: Comparison of TraVIS (single-scale GRU) across versus unimodal baselines. Best results are shown in bold.

Experiments	mAP@1s	mAP@2s
Visual branch only	68.85	71.74
Tracking branch only	68.05	69.51
TraVIS (single-scale GRU)	**75.27**	**77.35**

4.4.3 Tracking integration and multimodal fusion. **(a) Choice of temporal model for TraVIS.** Tab. 3 reports the performance of TraVIS using the two temporal modules selected in the preceding analysis. The single-scale bidirectional GRU variant of TraVIS achieves the best overall performance, reaching 75.27 mAP@1s and 77.35 mAP@2s. We attribute this to the multi-scale TCN suffering from training instability. We therefore adopt the single-scale GRU as the temporal module of TraVIS, as it offers higher performance together with a simpler and more stable architecture. **(b) TraVIS versus unimodal baselines.** Tab. 4 compares the best multimodal model, TraVIS (single-scale GRU), with the unimodal alternatives, showing it outperforms both (vision- and tracking-only). The former improves mAP@1s by 6.42 points over the visual stream and by 7.22 points over the tracking stream, proving the two modalities provide complementary information. We hypothesize that tracking contributes with structured spatial cues that may be difficult to infer from video alone, whereas visual features retain semantic information that cannot be recovered from trajectories alone. Their fusion consequently yields a more complete representation of the actions. **(c) Ablation study.** Tab. 6 presents ablations of key components of TraVIS. Removing *auxiliary supervision* reduces mAP@1s by 8.40 points, bringing performance close to that of the unimodal baselines. This suggests that, without explicit supervision for each modality, joint training may cause the model to rely predominantly on one stream rather than fully exploiting their complementary information. Second, removing *temporal self-attention* from the tracking stream decreases mAP@1s by 1.93 points. This indicates that modeling temporal dependencies also for the tracking modality before fusion eases the later joint temporal modeling. Finally, we evaluate an *alternative fusion strategy* based on feature concatenation followed by a linear projection. However, the simpler element-wise max-pooling operation performs slightly better, suggesting that it provides an effective and sufficiently expressive mechanism for combining the two modalities. **(d) Effect of graph connectivity.**

Table 5: Per-class AP@1s and overall mAP@1s for TraVIS compared to the vision-only baseline. The number of training instances is reported for each class, with classes ordered by frequency and the best result in each column shown in bold.

		Action class										
		Reception	Pass	Tackle	Throw-in	Shot at goal	Cross	Free-kick	Goal-kick	Corner	Kick-off	All
Train instances		9,054	9,018	1,718	344	267	255	220	144	81	46	21,147
AP@1s	Vision baseline	79.77	84.95	36.06	76.79	67.35	33.49	58.89	63.94	**87.69**	**99.52**	68.85
	TraVIS	**84.94**	**86.62**	**37.15**	**92.75**	**72.69**	**43.85**	**74.33**	**74.55**	87.67	98.18	**75.27**

Table 6: Ablation of different design choices for TraVIS (single-scale GRU). related to supervision, temporal self-attention for modeling frame-level GIN representations, and multi-modal fusion. Best results are shown in bold.

Experiments	mAP@1s	mAP@2s
TraVIS (single-scale GRU)	**75.27**	**77.35**
w/o auxiliary supervision	66.87	68.26
w/o temporal self-attention	73.34	74.74
Concatenation multimodal fusion	74.62	76.34

Table 7: Effect of graph connectivity on the tracking-only stream.

Experiments	mAP@1s	mAP@2s
Tracking only stream (proposed connectivity)	**68.05**	**69.51**
fully connected graph	65.15	66.48
w/o tactical edges	59.54	60.82
w/o proximity edges	65.41	66.41
w/o tactical or proximity edges	12.21	16.92

Tab. 7 evaluates different graph structures for the tracking-only stream. Our proposed connectivity, combining *ball-interaction, tactical, and proximity edges*, achieves the best performance. Removing edge types reduces accuracy, especially when tactical edges are excluded. Using only *ball-interaction edges* severely limits connectivity, as player-player interactions must be mediated through the ball node, preventing effective learning. Although a fully connected graph performs reasonably well, it remains inferior to the proposed structured design. These results show that semantically meaningful interactions are more effective than indiscriminate connectivity while also reducing computational cost.

4.5 Final results and per-class analysis

The final TraVIS model achieves 75.27 mAP@1s and 77.35 mAP@2s, demonstrating strong overall performance. Tab. 5 reports the corresponding per-class results. Performance is highest for *kick-off*, *throw-in*, and *corner*, achieving AP@1s values above 87, likely due to their distinctive spatial and visual configurations, such as restarting play from the center of the pitch, a corner, or the touchline. The frequent *pass* and *reception* classes also perform strongly, ar around 85 AP@1s. Performance is slightly lower for *goal-kick*, *free-kick*, and *shot at goal*, although all three remain close to 75 AP@1s. These classes exhibit greater intra-class variability, as they may occur from different locations and under diverse game situations. Finally, *cross* and *tackle* are the most challenging classes. We attribute their lower performance to their high visual and temporal variability and their similarity to other actions, which can make their temporal boundaries and class identity inherently ambiguous. Compared with the vision-only baseline, TraVIS improves performance for nearly all action classes. The largest gains are observed for *throw-in* and *free-kick*. For throw-ins, the explicit player and ball coordinates help identify restarts occurring near the pitch boundaries. For free-kicks, the tracking representation captures characteristic motion patterns, such as the ball remaining stationary before play resumes. These results indicate that structured tracking information is particularly valuable for actions defined by distinctive spatial configurations or player–ball dynamics.

5 Conclusion

In this work, we conducted a systematic comparative analysis of temporal modeling strategies for action spotting in football. We evaluated both single-scale and multi-scale approaches, showing that a single-layer bidirectional GRU provides a strong and computationally efficient baseline, particularly for short-duration actions that require limited temporal context. We also introduced TraVIS, a multimodal action spotting framework that augments visual features with structured player and ball tracking information. By incorporating football-specific inductive biases through tracking data, TraVIS improves upon the vision-only baseline by 6.42 mAP@1s points, demonstrating the substantial value of tracking information for football action spotting. Furthermore, our analysis highlights the importance of auxiliary, branch-specific supervision for enabling effective fusion between the visual and tracking modalities. All experiments were conducted on *PanoTrack*, a dataset comprising continuous footage captured by fixed panoramic cameras. This setting avoids many of the challenges associated with standard broadcast videos, such as camera cuts and viewpoint changes.

Limitations and future work. Although TraVIS achieves strong performance on our proprietary dataset, its generalization to larger datasets, different recording conditions, alternative pitch configurations, and other competitions remains to be evaluated. Extending the approach to other sports also requires adapting the graph representation to their specific structural characteristics. Future work will therefore assess the robustness and transferability of the proposed framework across datasets, acquisition settings, and sports.

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